

Consolidated VLA results (HT campaign) — single source for the paper tables

Detail files: vla_results.md (GR1 curves/per-task, pi0.5 per-suite), multimodality_table.md, memory notes nu-is-inert-heteroscedasticity-is-the-method.md. Updated 2026-08-27.

A. Cross-stack headline — single-pass HT vs flow / L1 / MSE (budget-matched, all entries annotated @steps)

Blank = not run. HT entries carry their nu. Within a row every number is at the same eval protocol; "@steps" is the training checkpoint evaluated.

stack (chunk d)	protocol	flow	L1	MSE	HT incumbent (nu = 2d)	HT swept-best	HT fixed c = 2 (nu = 4d)
GR00T GR1, robocasa (232)	24 tasks x 20 eps, official harness	44.1 @60k (pub 44.5 @60k)	17.2 @60k	37.8 @60k	nu=464: 47.5 @60k	nu=928: 51.5 @60k (nu=2: 24.5; nu=0.5: 44.8; nu=464: 47.5)	nu=928: 51.5 @60k
GR00T WidowX, bridge (56)	7 tasks x 50 eps (effective 10 scenes/task)	57.1 @20k			nu=112: 62.9 @20k	nu=224: 67.4 @20k (flat 49-67 over nu=2..448)	nu=224: 67.4 @20k
OpenVLA-OFT libero-long (56)	10 tasks x 10 eps (released ref: 500 eps, best ckpt)		released 94.5 (best ckpt <=150k)		nu=112: 82 @120k (arm endpoint)	nu=1024: 98 @150k	nu=224: 95 @150k
OpenVLA-OFT libero-spatial	10 tasks x 50 eps (500)		released 97.6		nu=112: 98.4 @50k (492/500)		nu=224: in training (150k)
OpenVLA-OFT libero-goal	same		released 97.9		nu=112: 98 @90k (97.6 @30k, 500 eps)	nu=1024: 98 @150k	
OpenVLA-OFT libero-object	same		released 98.4		nu=112: 95 (best ckpt)	nu=1024: 97 @150k	
pi0.5 LIBERO (350), 4-suite avg	4 x 1000 eps	published ~96.9		96.8 @30k	nu=700: 97.6 @30k (spatial 98.1 / goal 97.6 / object 99.8 / long 94.7)	(nu=2 97.0 @30k was a 1-GPU run, 8x fewer samples: not comparable)	nu=1400: 8-GPU rerun in training (a 1-GPU run gave 96.1: 97.8 / 95.8 / 98.6 / 92.2, not comparable)
Cosmos3 libero-10 (160)	10 tasks x 10 eps, 30-step sampler	96 @2000			nu=320: 93 @2000	nu=512: 96 @2000	nu=640: 96 @2000

Reading across rows: single-pass HT matches or beats its iterative baseline on every stack once nu is at a light-but-finite operating point (GR1 51.5 vs 44.1; WidowX 62.9 vs 57.1; OFT-long 98 vs released 94.5; pi0.5 97.6 vs ~96.9; Cosmos3 96 = 96). The accidental default nu = 2d underperforms only where the hard-but-learnable residual band matters (OFT-long 82, Cosmos3 93). The fixed prescription c = 2 (nu = 4d, the same rule on every stack) matches the swept optimum where it has been run to its endpoint (WidowX 67.4 = sweep best; Cosmos3 96 = flow; GR1 51.5 > the swept-at-2d incumbent 47.5, single seed, SE ~2.3), is within 3 points on OFT-long (95 vs 98 for nu=1024 at 150k; above the released L1 94.5). On pi0.5 the c = 2 cell is being re-run: the first attempt (and the earlier nu=2 point) had trained on 8x fewer samples than the incumbent (single-process launch), so those numbers are not comparable.

Huber-gate controls (budget-matched, not a table column): GR1 simple Huber (MSE inside 2.5 x EMA-median chunk-residual norm, linear beyond; no sigma) 37.4 @60k = MSE 37.8, vs HT 47.5 / flow 44.1; GR1 heteroscedastic Huber (Huber rho on ||r||/sigma + d log sigma) 40.7 @60k: sigma normalization recovers about half of the MSE->HT gap, the redescending gate the rest. OFT-long simple / hetero Huber 97 / 95 @50k (arms stopped at 80k). The cap reproduces HT on OFT-long but not on GR1.

Budget-mismatched controls (not in the table): OFT-long hetero-Gaussian 73 @50k (arm stopped at ~91k; 59 @20k); OFT-goal/object HG 94/88 @50k; GR1 pure HG 20.6 @22k and HG-pre->HT 32.8 @22k (HT @60k); in-house OFT L1 66 @20k and MSE 84 @20k. Arm status (2026-08-27). Completed: OFT-long fixed nu=224 (89 @20k, 94 @30k, 94 @50k, 95 @150k), Cosmos3 fixed nu=640 (62/83/97/96 @iter 500/1000/1500/2000), GR1 simple / hetero Huber (37.4 / 40.7 @60k), OFT-long learned-nu (93 @150k). Stopped before their endpoint (last evaluated snapshot is final): OFT-long median-pinned c=2.5 (95 @50k, stopped at 130k) and c=2 (88 @50k, at 60k), per-sample / global argmax-nu table (92 / 91 @50k, at 80k), OFT-long simple / hetero Huber (97 / 95 @50k, at 80k), Cosmos3 median-pinned c=2 (never evaluated). Completed for the c = 2 column: GR1 nu=928 51.5 @60k. pi0.5 nu=1400: 8-GPU rerun in training (run_c2x8). In training: OFT libero-spatial nu=224 (150k; 500-episode eval armed); queued for the next free nodes: OFT libero-goal and libero-object nu=224, pi0.5 nu=1400

(30k). DROID + Cosmos3 flow/HT (nu=1024) pair: data, recipe and smoke test done (4 samples/rank fit; 108 s per 512-sample iteration on one A800 node, i.e. ~6 days for 5k iterations), paused until the c = 2 cells are filled; no RoboLab evaluation stack exists yet. Full trajectories: memory note nu-is-inert-heteroscedasticity-is-the-method.md and vla_results.md.

B. Single-pass vs iterative cost

- HT is 1 NFE on every stack. Iterative baselines: GR1 flow 4 NFE, pi0.5 flow 10, Cosmos3 30. Cosmos3's HT is served through the same 30-step joint video/action sampler, so it is not a latency win there.
- OpenVLA-OFT's diffusion head at T_test = 1 collapses to 0.0 SR (their Table II): single-pass by construction is not free for iterative heads.

C. Mechanism results (mechanism section)

- Gradient share of the top-1% residual samples: HG 18.5% vs HT (nu=1024) 0.9% on OFT (per-sample sigma); pi0.5 9.4% -> 1.0%. Data/gradient proportion tables in the memory note.
- GR1 HG-vs-HT training-set fitting probe: HG fits worse on every channel; worst on the binary hand-transition channels (HG/HT rms ratio 1.44 vs 1.25 elsewhere, dim 24 = 1.89).
- Multimodality: 0 alternative-action states in 3000 screened states per stack; all above-chance sampling structure is gripper/hand event timing (fig_flow_unimodal.png, multimodality_table.md).
- Flow vs HG vs HT over the surprise tail (E7; same samples, flow checkpoints): HG AMPLIFIES the tail, flow is PROPORTIONAL to it (gradient share ~ data share; the (t, noise) draw dominates per-sample gradient, corr with the sample's residual ~0), HT SUPPRESSES it:

stack	tail bucket	data %	HG grad %	flow grad %	HT grad %
GR1 (flow ckpt @60k)	1-2x rms	23.4	40.0	28.0	27.0
GR1	2-4x rms	0.6	2.8	1.2	0.6
GR1	top 1% of samples	1.0	4.0	2.9	1.1
pi0.5 (flow ckpt libero_ft)	1-2x rms	36.0	62.4	36.6	53.3
pi0.5	2-4x rms	3.0	15.8	2.4	3.7
pi0.5	top 1% of samples	1.0	6.2	4.0	1.5

Matches the SR ordering HT >= flow > HG (GR1 47.5 / 44.1 / 20.6; OFT-long 98 / -- / 73).

- ML nu on converged residuals is 2.0-24 on every stack while SR-optimal nu is 224-1024: likelihood-optimal and learning-optimal gates are different objects.

D. Caveats to carry into the paper

- OFT cells: 100 episodes, single seed; the released reference is best-of-checkpoints over 500 eps.
- WidowX: 5 parallel envs share initial states -> effective n = 10 scenes per task (SE ~5.9 pts pooled).
- GR1 HG (22k) vs HT (60k): HG stopped early; the per-dimension structure of the fit gap is the step-robust signal, the uniform part is partly budget.
- pi0.5 "ref flow ckpt" numbers are our-harness evals of finetuned_v044 and sit well below the published figures; quote the published number as the baseline.
- GR1 HT best full-decay endpoint is 47.4 at a 50k schedule (no matched flow run at 50k); the strictly matched comparison is the 60k endpoint.

E. Gradient proportion tables (why HT > HG at large nu)

Method: per-sample residuals from several hundred training samples at each stack's trained HT checkpoint (dumps resid_dist_{oft,pi05,gr1}.npz; OFT = HTpair-long 120k, pi0.5 = run_ht 30k, GR1 = ft_ht3 60k). "data %" = share of samples in the bucket (error size in multiples of the pooled rms); "grad %" = the bucket's share of total squared-gradient mass, HG mass = S/σ^4 , HT mass = $w^2 * S/\sigma^4$ with $w = (nu+d)/(nu + S/\sigma^2)$. Measured at converged, gate-trained checkpoints, so the tails are as small as they ever get: these are lower bounds on what HG suffers during training.

E1. OpenVLA-OFT libero-long (960 samples, d=56, nu=1024, per-sample sigma = the loss's actual gate)

error size	data %	HG grad %	HT grad %
< 0.5x rms	46.0	25.7	44.0
0.5-2x (typical)	49.2	51.2	53.9
2-4x	4.0	17.3	1.8
4-10x	0.8	5.9	0.3

error size	data %	HG grad %	HT grad %
> 10x	0.0	0.0	0.0
top 1% of samples	1.0	18.5	0.9

E2. pi0.5 LIBERO (600 samples, d=350, nu=700, pooled sigma)

error size	data %	HG grad %	HT grad %
< 0.5x	24.5	2.9	7.7
0.5-2x	72.8	79.6	89.3
2-4x	2.7	17.5	3.1
4-10x	0.0	0.0	0.0
top 1%	1.0	9.4	1.0

E3. GR00T GR1 robocasa (320 samples, d=232, nu=464, pooled sigma)

error size	data %	HG grad %	HT grad %
< 0.5x	0.0	0.0	0.0
0.5-2x	99.4	97.4	99.4
2-4x	0.6	2.7	0.6
top 1%	1.0	3.9	1.0

GR1 note: chunk-level ratios are compressed by averaging over 232 dims (max 2.1x rms), so at pooled sigma the losses barely differ; GR1's spreading happens through its per-sample sigma head (median 0.035 vs pooled rms 0.199; 97.5% of samples past the gate knee under the trained sigma). The per-sample-sigma version would look like E1.

E4. Gradient pull per sample vs error size (OFT, sigma co-adapted to each nu; typical sample = 1)

loss	sigma / rms	gate knee	0.5x	1x	2x	3x	5x	10x
HG (nu=inf)	1.00	--	0.50	1.00	2.00	3.00	5.00	10.00
nu=1024 (swept best)	0.87	3.7x rms	0.53	1.00	1.66	1.94	1.90	1.29
nu=112 (incumbent)	0.63	0.9x rms	0.86	1.00	0.75	0.55	0.35	0.18
nu=2.2 (ML on residuals)	0.45	0.09x rms	1.95	1.00	0.50	0.34	0.20	0.10

Pull = gate weight x residual. Large nu keeps the rising (MSE-like) pull through the learnable 2-4x band and caps only the far tail; heavy nu inverts the curve (most pull to already-fit samples); HG's pull grows without bound.

E5. ML-optimal nu on the same converged residuals (the objective-mismatch result)

stack	ML nu (one fitted sigma)	ML nu (head's per-sample sigma)	SR-optimal nu	frac past knee at ML nu / at SR-optimal nu
OFT (d=56)	2.2	2.2	1024	100% / 1.3%
pi0.5 (d=350)	2.0	--	700	99.8% / 24%
GR1 (d=232)	23.6	--	464	100% / 3.7%

Maximum likelihood in nu is a fit criterion (it explains the over-dispersed normalized residuals with tails, halving sigma); the SR-optimal nu is a control criterion (keep MSE pull through the learnable band, gate only surprises). A likelihood-fitted nu therefore lands heavy.

E6. Gate scale: theory vs real training, with the swept optimum per stack

Knee in sigma units = $\sqrt{\text{nu}/d}$. Theory row assumes a calibrated sigma and Gaussian residuals ($u \sim \chi^2_d$). "past knee" = measured fraction of training samples with $S/\sigma^2 > \text{nu}$ at the listed checkpoint (per-sample head sigma for OFT/GR1/WidowX, pooled sigma for pi0.5).

stack (d)	setting	nu	k = nu/d	nominal knee	past knee (theory, Gaussian)	past knee (measured)
OFT (56)	incumbent 2d	112	2	1.41 sigma	~1e-5	61%
OFT	fixed c=2 (4d)	224	4	2.0 sigma	6e-22	29%

stack (d)	setting	nu	k = nu/d	nominal knee	past knee (theory, Gaussian)	past knee (measured)
OFT	swept best	1024	18.3	4.3 sigma	~0	3.6%
GR1 (232)	incumbent 2d = swept best	464	2	1.41 sigma	~1e-12	97.5% (sigma under-predicts 5.5x; Cauchy-like regime)
GR1	fixed c=2 (4d)	928	4	2.0 sigma	6e-84	90%
WidowX (56)	incumbent 2d	112	2	1.41 sigma	~1e-5	12.8% (gate probe @20k)
WidowX	swept best (flat axis)	224	4	2.0 sigma	6e-22	not measured
pi0.5 (350)	incumbent 2d = swept-equivalent (flat)	700	2	1.41 sigma	~1e-19	24% (pooled sigma)
Cosmos3 (160)	incumbent 2d	320	2	1.41 sigma	~1e-9	not measured
Cosmos3	swept best	512	3.2	1.79 sigma	~1e-24	not measured
Cosmos3	fixed c=2 (4d)	640	4	2.0 sigma	1e-58	not measured (SR 96 @2000)

Reading: the swept-optimal k is not stack-invariant (2 to 18, a 9x spread) because sigma's calibration is not; in realized terms the SR-optimal gate touches a few percent of samples on OFT but ~all samples on GR1 (whose optimum operates as a Cauchy-like loss).

E7. Flow-matching gradient distribution (does flow's training gradient concentrate on the tail?)

Method: per training sample, the flow-matching loss is measured with 8 Monte-Carlo draws of (t, noise) on the FLOW checkpoint, and the sample's action-space residual comes from the flow model's own sampled action (eval-setting sampler) vs ground truth. Flow's loss is plain MSE on the velocity residual, so a sample's gradient mass is proportional to its per-sample flow loss. HG/HT columns are computed on the same samples' action residuals (HG mass $\propto S$; HT mass $\propto w^2 S$ at the stack's incumbent nu, pooled sigma).

E8a. GR00T GR1 robocasa (320 samples, ft_flow @60k, 4-step sampler; d=232)

bucket (flow model's action residual)	data %	FLOW grad %	HG grad %	HT (nu=464) grad %
0.5-1x rms	75.9	70.7	57.2	72.4
1-2x	23.4	28.0	40.0	27.0
2-4x	0.6	1.2	2.8	0.6
top 1% of samples	1.0	2.9	4.0	1.1

Diagnostics: corr(per-sample flow loss, action residual) = 0.25; per-sample flow loss varies across the 8 (t, noise) draws with median CV = 0.95 — i.e., most of a sample's flow gradient is set by the random (t, noise) draw rather than by the sample's own residual. Flow's gradient distribution on GR1 sits between HG and HT: it does not concentrate on the tail like HG (the t/noise averaging dilutes per-sample differences) but it also does not gate; note GR1's chunk-level residuals have no far tail at pooled sigma (max ~2.1x rms), so this is the mild-tail case.

E8a-T. The same table across training (GR00T GR1, 320 samples, 8 (t, noise) draws, 4-step sampler; buckets relative to each checkpoint's own pooled rms)

checkpoint	pooled rms	bucket	data %	FLOW grad %	HG grad %	HT (nu=464) grad %
2k (ft_flow2k, SR 1.5%)	0.276	0.5-1x	69.7	62.0	40.8	61.9
		1-2x	30.0	37.7	57.9	37.7
		2-4x	0.3	0.2	1.3	0.4
		top 1% of samples	1.2	1.6	4.6	1.5
8k (ft_flow2k)	0.261	0.5-1x	70.6	65.7	44.9	64.1
		1-2x	29.1	33.5	53.6	35.6
		2-4x	0.3	0.7	1.5	0.3
		top 1% of samples	1.2	1.9	4.5	1.4
46k (keep/flow_46000)	0.217	0.5-1x	75.6	70.3	56.3	71.9
		1-2x	23.8	28.5	40.8	27.4
		2-4x	0.6	1.2	2.9	0.6
		top 1% of samples	1.2	2.1	5.2	1.3

checkpoint	pooled rms	bucket	data %	FLOW grad %	HG grad %	HT (nu=464) grad %
60k (ft_flow, converged; = E8a)	0.214	0.5-1x	75.9	70.7	57.2	72.4
		1-2x	23.4	28.0	40.0	27.0
		2-4x	0.6	1.2	2.8	0.6
		top 1% of samples	1.2	1.9	5.2	1.3

Diagnostics per checkpoint: 2k (ft_flow2k, SR 1.5%): rms 0.276, max residual 2.03x rms, corr(flow loss, residual) 0.46, median CV of flow loss across draws 0.66, flow loss median 0.1725; 8k (ft_flow2k): rms 0.261, max residual 2.18x rms, corr(flow loss, residual) 0.31, median CV of flow loss across draws 0.76, flow loss median 0.1885; 46k (keep/flow_46000): rms 0.217, max residual 2.17x rms, corr(flow loss, residual) 0.27, median CV of flow loss across draws 0.94, flow loss median 0.1939; 60k (ft_flow, converged; = E8a): rms 0.214, max residual 2.11x rms, corr(flow loss, residual) 0.27, median CV of flow loss across draws 0.95, flow loss median 0.1929.

Reading: the three regimes are already present at 2k and do not change in kind with training - HG's share of the 1-2x band and of the top 1% is above the data share at every stage, HT's is at or below it, flow's sits between (flow's top-1% share stays within 1-3x its data share at every stage, HG's within 2-5x). What changes with training is the residual scale (pooled rms falls) and the tail's extent; the ordering HG > flow > HT on the tail is stage-independent. No GR1 flow checkpoint exists between 8k and 46k (the ft_flow run kept only its last five), so the mid-early part of the curve is not covered; pi0.5 has no early/mid flow checkpoints at all (the flow baseline is the released endpoint).

E8b. pi0.5 LIBERO (600 samples, libero_ft flow ckpt, Beta-sampled t, 8 (t, noise) draws; 10-step sampler for the action residual; d=350)

bucket (flow model's action residual)	data %	FLOW grad %	HG grad %	HT (nu=700) grad %
< 0.5x rms	26.2	28.1	3.6	9.5
0.5-1x	34.8	32.9	18.2	33.5
1-2x	36.0	36.6	62.4	53.3
2-4x	3.0	2.4	15.8	3.7
top 1% of samples	1.0	4.0 (single draw 5.8)	6.2	1.5

Diagnostics: corr(per-sample flow loss, action residual) = -0.03 (none); per-sample flow loss CV across draws = 0.63. The flow gradient distribution tracks the DATA distribution almost exactly (28/33/37/2.4 vs 26/35/36/3.0): flow's per-sample gradient is set by the random (t, noise) draw and is essentially blind to whether the sample is a surprise. It neither chases the tail (HG puts 78% of its gradient on the >1x band, 16% on the 2-4x surprise band) nor gates it (HT: 4%); the tail's share under flow equals its data share. Probe pitfall (recorded): the residual scaffold's sample_time -> ones override must be removed for a flow-loss measurement — with t=1 the loss collapses to the single-pass residual and all draws coincide.

Summary of E7: three distinct gradient regimes over the surprise tail — HG amplifies it (share >> data share), flow is proportional to it (share ~ data share; the (t, noise) noise floor dominates per-sample differences), HT suppresses it (share << data share). This is consistent with the success-rate ordering HT >= flow > HG on GR1 and OFT-long.